

PRUDENCE ISLAND POTTER COVE, RI, USA

WMO: 997294

Lat: 41.637N

Lon: 71.339W

Elev: 3

StdP: 101.29

Time Zone: -5.00 (NAE)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -11.5	(c) -9.3	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 12.1	(k) -1.7	(l) 10.8	(m) -2.6	(n) 4.2	(o) 320	(p) 0.444

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 7.1	(c) 30.0	(d) N/A	(e) 28.4	(f) N/A	(g) 27.1	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 2.4	(p) 200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 10.9	(b) 8.9	(c) 7.6	(d) DB	(e) -14.1	(f) 33.2	(g) 2.4	(h) 2.1	(i) -15.9	(j) 34.7	(k) -17.3	(l) 35.9	(m) -18.6	(n) 37.0	(o) -20.4	(p) 38.5	(q) N/A
(4)			(r) WB	(s) N/A	(t) N/A	(u) N/A	(v) N/A	(w) N/A	(x) N/A	(y) N/A	(z) N/A	(aa) N/A	(ab) N/A	(ac) N/A	(ad) N/A	(ae) N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.5	-0.7	0.3	3.8	9.7	14.7	19.5	23.4	22.6	19.2	13.7	7.8	3.4		
	DBStd	9.04	5.04	4.38	4.12	3.30	3.27	3.11	2.34	2.14	2.84	3.60	4.18	4.56		
	HDD10.0	1154	331	272	195	44	2	0	0	0	0	10	91	209		
	HDD18.3	2898	589	506	450	260	120	22	1	23	147	317	463			
	CDD10.0	1708	0	0	3	35	148	285	416	391	276	126	23	4		
	CDD18.3	410	0	0	0	1	8	57	158	133	49	4	0	0		
	CDH23.3	1942	0	0	0	9	37	262	877	611	134	13	0	0		
	CDH26.7	383	0	0	0	2	7	58	204	97	12	2	0	0		
(14) Wind	WSAvg	3.3	3.6	3.8	4.1	3.8	3.3	2.7	2.4	2.4	2.9	3.5	3.8	3.7		
(15) Precipitation	PrecAvg	1176	93	88	120	105	88	91	84	96	102	99	99	111		
	PrecMax	1425	157	165	387	204	188	271	257	199	235	330	171	183		
	PrecMin	942	42	45	7	39	29	0	25	20	21	9	19	32		
	PrecStd	135	31	34	77	43	34	67	45	46	49	66	40	39		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.4	11.8	17.7	23.9	27.4	31.3	32.7	30.9	28.2	25.1	18.5	15.8		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	DB	10.7	9.5	13.5	19.1	23.7	28.2	30.6	28.9	26.2	21.7	16.5	13.6		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	DB	8.8	7.8	11.2	16.5	21.8	26.2	28.9	27.7	24.9	20.1	15.1	12.2		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	DB	6.8	6.2	9.3	14.8	20.0	24.5	27.4	26.5	23.7	18.8	14.1	10.6		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(35) Mean Daily Temperature Range	MDBR		7.0	7.2	7.3	8.0	7.9	7.4	7.1	7.2	7.1	6.9	6.5	6.4		
	5% DB	MCDBR	8.7	8.5	9.3	10.8	10.2	9.9	8.9	8.2	7.7	7.8	7.1	8.0		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(40) Clear-Sky Solar Irradiance	taub		0.302	0.311	0.335	0.374	0.416	0.461	0.486	0.443	0.375	0.342	0.314	0.302		
	taud		2.502	2.481	2.460	2.383	2.299	2.209	2.150	2.261	2.424	2.525	2.562	2.559		
	Ebn at Noon		854	902	915	897	862	820	795	820	859	854	836	828		
	Edh at Noon		73	87	100	116	129	142	149	129	102	83	69	65		
(44) All-Sky Solar Radiation	RadAvg		1.86	2.74	3.75	4.83	5.35	5.84	6.02	5.41	4.40	2.97	2.03	1.56		
	RadStd		0.14	0.26	0.27	0.52	0.53	0.58	0.32	0.28	0.32	0.27	0.15	0.15		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (66 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+95	+47

Nomenclature: See separate page